Thermic radiation of Leslie's cube and a black body

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A Leslie's cube and a cavity radiator were separately used to verify the T^4 dependence predicted by the Stefan-Boltzmann law. The infrared transmission of four different filters was examined. These included a window glass (SiO_2) filter with a radiation transmittance of $(1.31 \pm 0.63)\,\%$, silicon with $(46.38 \pm 0.64)\,\%$ and , NaCl with $(74.36 \pm 0.65)\,\%$. The angular dependence predicted by Lambert's law was tested for accuracy using the cavity radiator and, with the employed method, was approximately confirmed to hold.

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1 Introduction

Every object with a temperature above absolute zero emits electromagnetic radiation, known as thermal radiation. The intensity and spectral distribution of this radiation strongly depend on the temperature and surface properties of the material.

A major breakthrough in the theoretical description of thermal radiation was achieved by Max Planck in 1900, when he introduced the concept of quantized energy exchange. His work showed that energy can only be emitted or absorbed in discrete packets, called quanta, laying the foundation of quantum physics. The formula he derived describes the spectral energy distribution of an ideal blackbody.

To approximate blackbody behavior in experiments, a cavity with a small opening is commonly used. Additionally, a Leslie cube—filled with water at constant temperature and with sides of different surface finishes—can be used to compare the emissive properties of real materials.

The aim of this experiment is to analyze the radiative behavior of real surfaces as a function of temperature and surface characteristics. The Stefan-Boltzmann law, which states that the radiated energy of an ideal blackbody is proportional to the fourth power of its absolute temperature, is experimentally tested. Furthermore, the angular distribution of emitted radiation is studied based on Lambert's cosine law. Finally, the absorption of infrared radiation by different materials is examined.

2 Theoretical backround

In the following sections, the emission behavior of different surfaces and that of a cavity radiator will be examined. The ideal radiator is referred to as a blackbody, which is capable of completely absorbing and emitting the entire spectrum of electromagnetic radiation. Its spectral distribution follows Planck's law:

$$\rho(\nu, T)\, d\nu = \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{e^{\frac{h\nu}{k_B T}} - 1}\, d\nu \tag{1}$$

Other bodies are only "black" over certain frequency ranges, meaning they can absorb and emit radiation completely only within limited spectral intervals.

The total emitted power P of a blackbody is given by the Stefan-Boltzmann law:

$$P = \sigma A T^4 \tag{2}$$

where $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}K^{-4}}$ is the Stefan-Boltzmann constant and A is the emitting surface area.

The wavelength at which the radiation density reaches its maximum is described by Wien's displacement law:

$$\lambda_{\max} = \frac{2897.8\,\mu\mathrm{m\,K}}{T} \tag{3}$$

A black body emits radiation isotropically, meaning its radiance is independent of direction. When measuring the radiation on a surface at an angle ϕ, the effective area exposed to the radiation is reduced

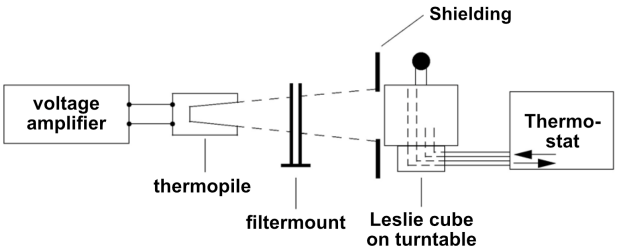

Figure 1: The thermostat regulates the temperature of Leslie's cube, which is rotatably mounted so that the four different sides can be examined. Its emitted heat radiation is measured via the thermic element, whose signal is amplified. A filter can be mounted in the radiation's path.[1]

by a factor of $\cos(\phi)$. Therefore, the measured radiance decreases according to Lambert's law:

$$J = J_0 \cos(\phi) \tag{4}$$

where J_0 is the radiance measured at normal incidence ($\phi = 0°$).

3 Emission behavior of different surfaces using the Leslie's Cube

3.1 Experimental setup

The experimental setup is shown in Fig. 1. A Leslie cube was used to investigate the dependence of thermal radiation on the surface properties and chemical composition of different materials. The cube is mounted on a rotatable base, allowing each side to be positioned toward the detector during measurement. The available surfaces were: one black, one white, and two nickel-plated sides — one matte and the other polished.

The cube was heated using a Lauda E100 thermostat, which allowed the temperature to be increased from 20 °C to 90 °C. The emitted thermal radiation was measured using a Kipp & Zonen thermopile. This sensor detects infrared radiation, and through the Seebeck effect, produces a voltage signal proportional to the radiation intensity. The resulting voltage was recorded using an Agilent 34405A digital multimeter.

3.2 Results

Due to the measurement principle, Equation 2 must be corrected by the reference temperature T_0, that is the temperature at which the output voltage of the thermopile is zero. It corresponds to the operating temperature of the thermopile. We measured it directly and got $T_0 = (24 \pm 1)$°C. With this correction, $U \sim \left(T^b - T_0^b\right)$ was used to fit the measured data according to

$$y = a * (T^b - T_0^b). \tag{5}$$

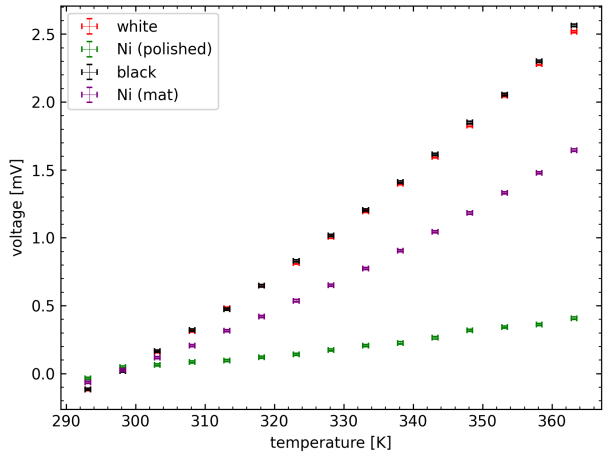

Figure 2: Measured voltage from the thermopile - which is proportional to the thermal power - as a function of surface temperature for all four sides of the Leslie cube. The surfaces are white (red), black (black), matte nickel-plated (purple), and polished nickel-plated (green).

Here, U is the voltage of the amplifiered signal of the thermopile which is proportional to the thermal power and T is the surface temperature. The exponent b is expected to be close to 4, as predicted by the Stefan-Boltzmann law (see Eq. 2).

Surface	**Fitted exponent** b
White	3.942 ± 0.066
Nickel (polished)	1.81 ± 0.99
Black	4.043 ± 0.069
Nickel (matte)	3.964 ± 0.051

Table 1: Determined exponent b for all four surfaces of the Leslie cube, obtained by fitting the model $a \cdot (T^b - T_0^b)$ to the measured data shown in Fig. 2.

Following values for the exponents shown in Tab. 1 were obtained.

The values observed for the black, white and mat surface agree, within the uncertainty range, with the expected exponent of $\alpha = 4$ according to the Stefan-Boltzmann law. The black and white surfaces appear to transmit thermal radiation similarly well, which can be attributed to the fact that they are made of the same material—the Leslie cube's base material. This result is somewhat counterintuitive, as one might expect the black surface to have a higher absorptivity than the white surface. In the visible spectral range, the white surface reflects nearly all incoming radiation, while the black surface absorbs it. However, the emissive and absorptive behavior in the visible spectrum does not necessarily apply in the infrared region, which is relevant here. The matte Nickel

surface appear to transmit thermal radiation similarly well, although it is made of a different material. That is because incoming radiation is scattered in many directions, reducing reflection. This increases the chance that photons are absorbed rather than reflected. Their rough texture causes multiple internal reflections, enhancing absorption.

The exponent determined for the polished nickel surface is significantly lower than those of the other surfaces. This can be attributed to the material properties of nickel. When an electromagnetic wave strikes one of the nickel surfaces, a node is formed in the electric field at the surface. This induces a displacement of electrons within the metal, which acts to counter the incident electric field. As a result, infrared radiation is reflected more strongly, and less radiation is transmitted. The thermopile therefore measures a lower thermovoltage than in the cases of the other surfaces.

A potential source of systematic error is the drift of the reference temperature T_0. Due to the lack of cooling on the far side of the thermopile, its temperature increases over time during the measurement. As a result, the measured thermovoltage decreases. Consequently, the fitted exponent is systematically lower than the theoretical value that would be obtained without this error.

3.3 Infrared permeability for several materials

In this experiment, the setup from Fig 1 was reused, but just with the black surface and different optical filters inserted between the Leslie Cube and the detector. The filters tested were: window glass, silicon, and sodium chloride.

At a temperature of 75 °C (with an ambient room temperature of approximately 26 °C), the thermoelectric voltage is first measured without any filter. Then, the ambient radiation is measured by inserting a nickel-plated filter in front of the thermopile, which does not transmit any radiation. Since this filter blocks all radiation from the Leslie cube, it allows measurement of the thermo-electric voltage due to the surrounding environment. The ambient radiation measurement can be affected by reflections from the walls opposite the Leslie cube. This procedure is repeated for the three other filters. The transmittance τ of these materials is calculated using the formula:

$$\tau = \frac{U_F - U_0}{U_{0F} - U_0}$$

where U_F is the voltage measured with the filter, U_0 the ambient voltage, and U_{0F} the voltage without any filter. It is expected that the filters exhibit material-dependent transmission behavior.

The results summarized in Table 2 clearly demonstrate that the transmission of electromagnetic radiation, particularly in the infrared range, is strongly dependent on the structure and properties of the filter material. Infrared radiation has energy comparable to the vibrational modes of molecular bonds. Absorption occurs when the incident radiation excites these vibrations. As different bond types require different excitation energies, transmission is highly material-specific.

In amorphous materials such as window glass the molecules in the material are excited into vibrational motion by incident infrared radiation. So a significant portion of the infrared radiation is absorbed through broad energy bands, resulting in low transmission. This is reflected in a very small transmission coefficient of

$$Q_{\text{window}} = 1.31\%.$$

The energy of infrared radiation is insufficient to overcome the band gaps of silicon 1.12 eV [2] and sodium chloride 7.7eV - 8.0 eV [3]. Consequently, the energy cannot be absorbed by bound electrons in these materials. Additionally structural properties of the materials also play an essential role. Both silicon and NaCl have a crystalline lattice structure with well-defined, periodic atomic arrangements and fixed interatomic bonding angles. Unlike amorphous window glass, which has only short-range atomic order (disordered structure), these materials exhibit long-range order. This rigid lattice reduces the ability of atoms to convert absorbed IR energy into lattice vibrations. As a result, crystalline structures tend to transmit more incoming infrared radiation, whereas amorphous materials, such as glass, can absorb more due to their more flexible atomic arrangement.

The low transmittance of electromagnetic waves in silicon

$$Q_{\text{silicon}} = 46.38\%$$

is primarily because the doped silicon can only absorb infrared light due to dopants.

Unlike in metals, where free electrons block high-frequency waves, NaCl has very low free-electron mobility. Therefore, its effective plasma frequency is much lower than that of a metal. If the frequency of the incident wave is higher than the plasma frequency, the electrons cannot oscillate fast enough to reflect the wave, allowing transmission. This is the case for sodium chloride in the infrared region, which explains its relatively high transparency of

$$Q_{\text{NaCl}} = 74.36\%$$

Finally, the corresponding wavelength of the intensity maxima at room temperature and at the temperature of the Leslie cube in this part of the experiment is determined using Wien's displacement law (see Eq. 3) and listed in Table 3.

material	infrared permeability (%)
window glass	1.31 ± 0.63
p-doped silicon	46.38 ± 0.64
NaCl	74.36 ± 0.65

Table 2: Transmittance of infrared radiation through various materials at a temperature of 75°C.

T [°C]	$\lambda[\mu m]$
26.1	9.683 ± 0.013
75.0	8.3233 ± 0.0096

Table 3: Wavelengths of the emission maxima

3.4 Black body radiation

The blackbody radiator consists of an electric oven in which a hollow cylinder is heated. The oven is powered by a variable transformer. The temperature is measured using a thermocouple (see Fig.3).

3.4.1 Stefan–Boltzmann law for cavity radiation

For this experiment, the orientation is 0°. During the continuous heating of the black body, the voltage on the multimeter Agilent 34405 A was read off in 5°C steps. The measured temperature dependence is shown in Fig. 4. The data was fitted according to the equation

$$y = a * (T^b - c^b)$$

. From this, the following exponent can be determined:

$$b = (3.8807 \pm 0.0089)$$

This result cannot be reconciled with the theoretical value within the error.

Due to the T_0 drift of the reference temperature, which was already discussed in the previous part of the experiment, a systematic error is also present here. The drift increases with rising temperatures in the cavity radiator. This could be the reason for the deviation from the theoretical value. Taking this into account, the Stefan-Boltzmann law fits the oven used, which can therefore be considered a good model of a black body radiator.

As can be seen from the plot in Fig. 5, a linear relationship between the voltage U — and therefore the radiated power —and T^4 is nevertheless plausible.

3.4.2 Lambert's cosine law

Lambert's law states: If a radiation source or reflected radiation exhibits a constant and direction-independent radiance over a surface, then the emitted radiant intensity decreases with decreasing angle φ according to Eq. 4.

To investigate this in more detail, the radiated power of the cavity radiator was measured as a function of the angle φ, ranging from $\varphi = 0°$ to $\varphi = 50°$.

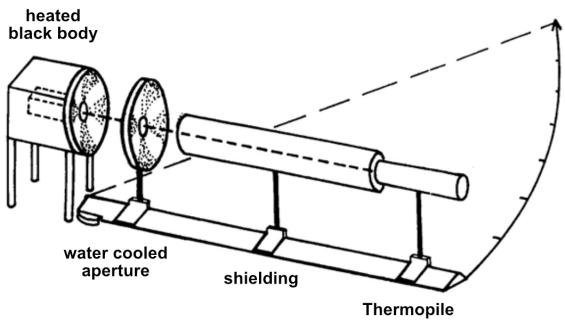

Figure 3: Experimental setup for measuring the temperature and angular dependence of an approximate blackbody using a cavity radiator.[1]

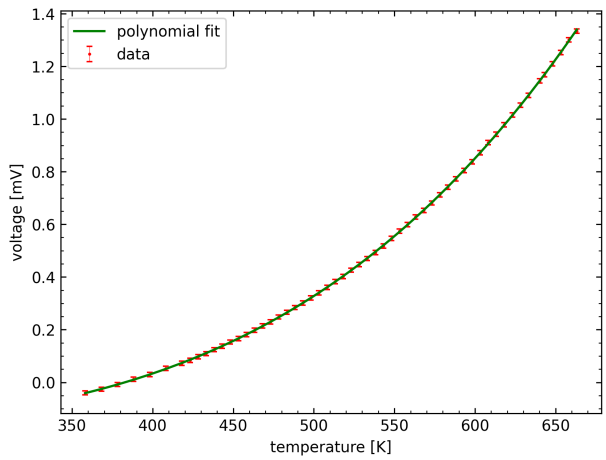

Figure 4: Voltage dependent on the temperature of a cavity radiator. Fit-function is $a*(T^b - c^b)$, $b = 3.8807 \pm 0.0089$, $c = 382.80 \pm 0.31$.

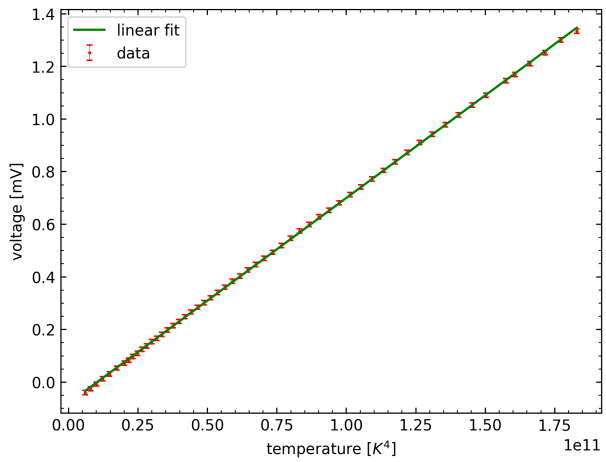

Figure 5: Voltage dependent on the temperature to the power of four of the cavity radiator and a linear fit according to $a * (T^4 - c^4)$, $a = (0.77987 \pm 0.00096)\frac{mV}{10^{11}K^4}$.

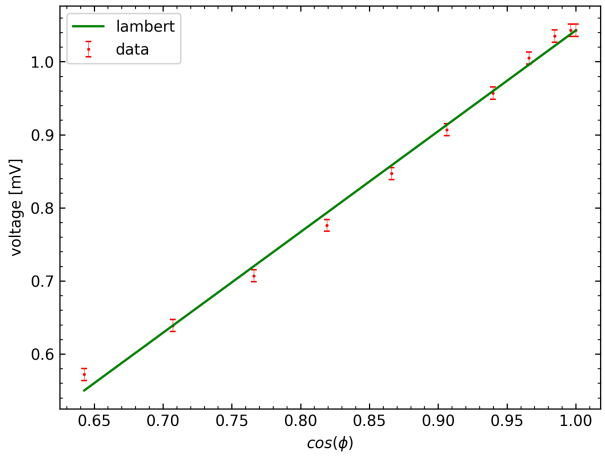

Figure 6: Thermoelectric voltage dependent on the cosine of the direction φ and a linear fit according to $a * cos(\phi) + b$, $a = (1.379 \pm 0.032)$.

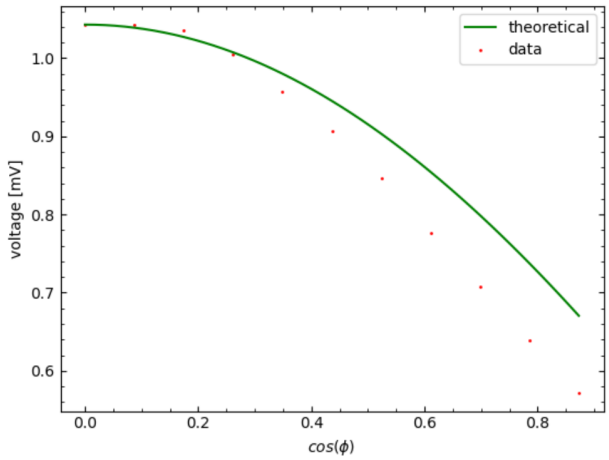

Figure 7: Plot of the thermoelectric voltage U and the theoretically expected function versus the angle φ. At larger angles, an increasing deviation of the measured values from the theoretical curve can be observed.

In Fig.6, the voltage measured by the thermopile is plotted against $\cos(\varphi)$ and fitted linearly. The slope of the linear fit in Fig.6 is $a = (1.379 \pm 0.032)$. This slope is not consistent with $a = 1$. The decrease in thermoelectric voltage with increasing angle φ originates from the orientation of the cavity radiator, which is directed toward $\varphi = 0°$. Additionally, the oven emits radiation preferentially in one direction, causing the thermoelectric voltage to drift downward at larger angles.

The aperture of the cavity radiator is positioned lower than the thermopile, which further distorts the measurements. The drop in normalized thermoelectric voltage at larger angles, as shown in Figs.7 can also be explained by the fact that background radiation is more influenced by the cavity radiator at small angles than at large ones. Furthermore, the temperature could not be held constant, as it was regulated via a current-driven variable transformer. As a result, the thermoelectric voltage U exhibited a relatively larger voltage difference at small angles compared to large ones, since the temperature steadily decreased during the measurement process.

This explains the larger slope of $a = (1.379 \pm 0.032)$. Despite these limitations, the applied method allows for an approximate confirmation of Lambert's law.

4 Conlusion

In the first part of our paper, we examine how well the radiation from the different surfaces of Leslie's cube follows the T^4 temperature dependence predicted by Stefan and Boltzmann.

The exponent for the black side of Leslie's cube is $b = 4.043 \pm 0.069$, the white side has $b = 3.942 \pm 0.066$, the matte side follows an exponent of $b = 3.964 \pm 0.051$, and the polished nickel-plated side has $b = 1.81 \pm 0.99$.

We therefore conclude that Stefan-Boltzmann's law can be used to predict the intensity of radiation of the tested materials except the polished Nickel surface. For materials like polished nickel, adjustments should be made if an exact power law is required.

In the next part, an oven with a larger temperature range was used to examine the T^4 power law. An exponent of $b = 3.8807 \pm 0.0089$ was found, which suggests Stefan-Boltzmann's law.

If different materials are placed in the path of the thermal radiation and the measured intensity is compared to that without any obstructions, we can determine the effectiveness of different materials as infrared filters. The wavelengths with maximal thermal radiation used in this experiment - calculated according to Wien's law - are $\lambda_{\text{room}} = (9.683 \pm 0.013)\,\text{m}$ to $\lambda_{\text{Leslie}} = (8.3233 \pm 0.0096)\,\text{m}$, and thus lie entirely within the infrared region of the electromagnetic spectrum.

The transmission values are as follows:

- Window glass: $(1.31 \pm 0.63)\%$
- Silicon: $(46.38 \pm 0.64)\%$
- NaCl: $(74.36 \pm 0.65)\%$

If the black body is not observed head-on, but at an angle, the intensity is expected to follow Lambert's cosine law. The radiation from the oven observed from above was found to approximately follow this law especially for smaller observation angles.

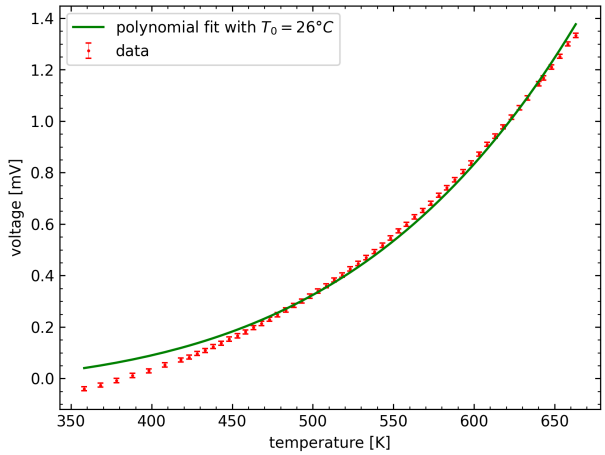

Figure 8: fit-function $a * (T^b - 299.15^b)$, $b = 4.889 \pm 0.084$, $299.15K = 26.0°C = T_R$

References

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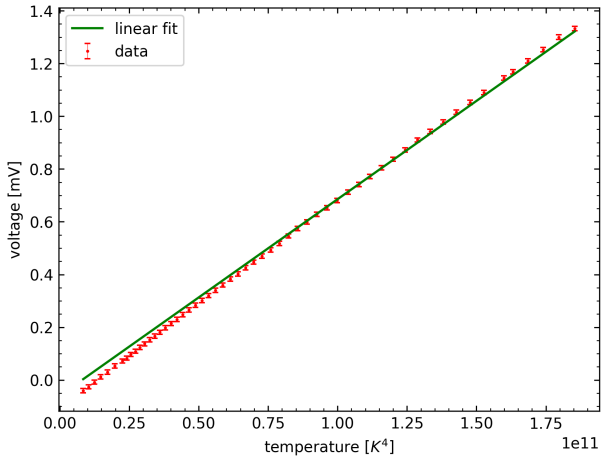

Figure 9: fit-function $a * (T^4 - 299.15^4)$, $299.15K = 26.0°C = T_R$

5 Appendix